NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Information Technology
Karachi, Pakistan

PROJECT REPORT

Intelligent Medical Diagnosis Expert System

Shifa • *Rule-based diagnosis. Every answer explained.*

Forward chaining • Backward chaining • MYCIN certainty factors

Course code	CT-361
Course title	Artificial Intelligence and Expert Systems (AIES)
Assessment	Complex Computing Problem (CCP)
Course teacher	Dr. Usman Amjad
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1. Introduction and Requirements

Shifa is a rule-based expert system for medical diagnosis. The user enters symptoms (buttons, or a typed sentence in English or Roman Urdu). The system returns a ranked list of possible diseases with a confidence level, the reasons, treatment and preventive measures. All medical knowledge is in one JSON file; the Python code has no disease names.

Table 1. Project at a glance.

Item	Value	Detail
Diseases	**6**	3 severe (pneumonia, dengue, malaria), 2 moderate (typhoid, influenza), 1 mild (common cold)
Symptoms	**20**	16 answered by intensity (mild / moderate / severe), 4 answered yes / no
Rules	**23**	20 diagnostic rules (R1–R20) and 3 red-flag rules (RF1–RF3), plus 7 intermediate facts
Reasoning	**2 modes**	Forward chaining always; backward chaining when two diseases are close
Uncertainty	**MYCIN CF**	Certainty factor 0 to 1 per disease, shown as confidence %
Interfaces	**CLI + GUI**	Command line and a Streamlit app (4-step wizard)
Testing	**132**	Automated checks, all pass (14 clinical cases, 7 regression, 8 hybrid cases)

1.1 Requirement coverage

Table 2. Every CCP requirement and deliverable, where it is covered, and the file that implements it.

CCP requirement	How we meet it	Section	File
Symptoms (min 8–10)	20 symptoms	3.1	`knowledge_base.json`
Diseases (min 3–4)	6 diseases	2	`knowledge_base.json`
Rules combining symptoms	20 AND-rules + 3 red flags	3.3–3.4	`knowledge_base.json`
Severity mild/moderate/severe	Per disease and per symptom	2, 3.2	`knowledge_base.json`
Treatments, preventive measures	Every disease has both	2	`knowledge_base.json`
Possible diseases from symptoms	Forward chaining	4.2	`engine.py`
Severe over mild	Priority + severity-first ranking	4.4	`engine.py`
Justification per diagnosis	Rule chain, why / why-not	4.5	`engine.py, cli.py`
Confidence levels (A 70%, B 30%)	Confidence % + evidence strength	4.6	`engine.py`
CLI or GUI	Both	5	`main.py, app.py`
Chosen diseases description	Tables with source	2	Report
Facts and rules used	All tables	3	Report
Inference reasoning explained	Traces, formulas, figures	4	Report
Example queries + screenshots	5 queries, 8 screenshots	5	Report
Assumptions and limitations	Tables	7	`LIMITATIONS.md`
Python source code	In the zip file	Appendix	Whole project
Title page	Course, title, members, roll no.	Page 1	Report

2. Chosen Diseases

Six diseases cover all three severity levels. Dengue and malaria share early symptoms (fever, headache, nausea), which tests backward chaining. Cough appears in three diseases with different strength, and typhoid needs fever for more than 3 days, which needs a 3-level rule chain. WHO and CDC publish a symptom list for each, so every rule has a source.

Table 3. The six diseases: description, treatment and prevention (knowledge base entries, shortened). Sources: WHO, CDC.

Disease	Severity	Cause and spread	Main symptoms	Treatment / recommendation	Prevention	Source
Pneumonia	Severe	Lung infection (bacteria, virus or fungus)	Cough, shortness of breath, fever, chest pain	See a doctor promptly. Antibiotics if bacterial (amoxicillin preferred by WHO), hospital care if severe, rest and fluids.	Vaccines (pneumococcal, Hib, measles, flu), handwashing, less indoor smoke, no smoking.	WHO
Dengue	Severe	Virus, spread by Aedes mosquitoes	High fever, pain behind the eyes, body aches, nausea, rash	No specific medicine. Rest, plenty of fluids, paracetamol. Do NOT take aspirin or ibuprofen (bleeding risk). See a doctor and watch for warning signs.	Mosquito repellent, long sleeves, window screens and nets, remove standing water.	WHO, CDC
Malaria	Severe	Plasmodium parasite, spread by Anopheles mosquitoes	Fever, headache, chills	Blood test to confirm. Artemisinin-based combination therapy (ACT) prescribed by a doctor. Always needs medicine; severe signs need emergency care.	Treated bed nets, repellent after dusk, protective clothing, preventive medicine for travellers, vaccines for young children in high-risk areas.	WHO
Typhoid	Moderate	Salmonella Typhi bacteria, spread by contaminated food and water	Prolonged high fever, abdominal pain, constipation or diarrhoea	Blood test to confirm. Antibiotics for the full course. Wash hands after the toilet; do not prepare food for others until cleared.	Safe (boiled) water, sanitation, well-cooked hot food, hand hygiene, typhoid vaccine.	WHO
Influenza	Moderate	Influenza virus, respiratory infection	Sudden fever, dry cough, headache, muscle and joint pain, tiredness	Rest, fluids, treat fever. Seek care if worse. Antivirals early for high-risk or severe cases (doctor-prescribed).	Yearly vaccine, handwashing, cover coughs and sneezes, stay home when ill.	WHO
Common Cold	Mild	Virus of the nose and throat	Sneezing, runny nose, sore throat; usually no fever	Rest and fluids; saline spray, humidifier, lozenges; honey for cough (age 1+). Antibiotics do not help. See a doctor if breathing is hard, fever lasts over 4 days, or symptoms last over 10 days.	Wash hands, avoid touching the face, avoid close contact with sick people.	CDC

Table 4. Severity levels and how the engine uses them.

Level	Meaning	Diseases	Rule priority
Severe	Can be life-threatening; needs urgent care.	Pneumonia, Dengue, Malaria	**30**
Moderate	Serious but treatable; complications if untreated.	Typhoid, Influenza	**20**
Mild	Usually clears without treatment.	Common cold	**10**

3. Facts and Rules Used

Knowledge was taken from 9 WHO and CDC pages (see References) and written as data in knowledge_base.json. Three kinds of facts are used: user facts (symptoms), intermediate facts (signs) and conclusions (diseases).

3.1 Symptoms (20 user facts)

Table 5. Symptoms with answer type and synonyms used by the text parser (* asked only after fever is reported).

#	Symptom	ID in file	Answer	Synonyms (English, Roman Urdu)
1	**Fever**	`fever`	intensity	temperature, high temperature, feverish, bukhar
2	**Fever lasting more than 3 days**	`fever_more_than_3_days`	yes / no *	prolonged fever, fever for days
3	**Chills**	`chills`	intensity	shivering, rigors, kapkapi
4	**Sweating**	`sweating`	intensity	night sweats, paseena
5	**Headache**	`headache`	intensity	head pain, sir dard
6	**Body / muscle / joint pain**	`body_aches`	intensity	muscle pain, joint pain, body pain, badan dard
7	**Fatigue**	`fatigue`	intensity	tiredness, weakness, malaise, kamzori
8	**Cough**	`cough`	intensity	dry cough, khansi
9	**Sore throat**	`sore_throat`	intensity	throat pain, gala kharab
10	**Runny nose**	`runny_nose`	intensity	nasal discharge, blocked nose, zukam
11	**Sneezing**	`sneezing`	intensity	sneeze, cheenk
12	**Skin rash**	`rash`	yes / no	red spots, skin spots
13	**Pain behind the eyes**	`pain_behind_eyes`	intensity	eye pain, retro-orbital pain
14	**Bleeding gums or nose**	`bleeding_gums`	yes / no	gum bleeding, nose bleed
15	**Nausea or vomiting**	`nausea_vomiting`	intensity	nausea, vomiting, ulti
16	**Abdominal pain**	`abdominal_pain`	intensity	stomach pain, belly pain, pait dard
17	**Diarrhoea or constipation**	`diarrhea_or_constipation`	intensity	loose motions, diarrhea, constipation
18	**Shortness of breath**	`shortness_of_breath`	intensity	breathlessness, difficulty breathing, saans phoolna
19	**Chest pain**	`chest_pain`	intensity	chest tightness, seene mein dard
20	**Confusion**	`confusion`	yes / no	disorientation, drowsiness, altered consciousness

3.2 Severity values and intermediate facts

Table 6. Symptom intensity becomes a certainty factor (CF).

User answer	CF	Meaning
Mild	**0.4**	weak evidence
Moderate (default)	**0.7**	usual evidence
Severe, or yes (yes/no symptom)	**1.0**	strongest evidence
No	**0**	absent; not passed to the engine

Table 7. Seven intermediate facts: conclusions that are not diseases but let rules chain.

Fact	Meaning	Produced by	Used by
Pneumonia signs	Evidence pattern consistent with pneumonia	R1, R2	R3
Dengue signs	Evidence pattern consistent with dengue	R4, R5, R6	R7
Malaria signs	Evidence pattern consistent with malaria	R8, R20, R9	R10
Typhoid signs	Evidence pattern consistent with typhoid	R12	R13
Prolonged fever	Fever persisting more than 3 days	R11	R13
Flu signs	Evidence pattern consistent with influenza	R14, R15	R16
Cold signs	Evidence pattern consistent with the common cold	R17, R18	R19

3.3 Diagnostic rules R1–R20 (heuristic knowledge)

Table 8. The 20 rules: IF (AND of conditions) THEN fact or disease, with certainty factor (CF) and priority (Pri.). Disease conclusions are bold.

Rule	IF	THEN	CF	Pri.	Reason / evidence
R1	Cough AND Shortness of breath	Pneumonia signs	0.80	30	Lower respiratory pattern
R2	Chest pain	Pneumonia signs	0.50	30	Chest pain listed for pneumonia
R3	Fever AND Pneumonia signs	**Pneumonia**	0.85	30	Fever + lower respiratory signs

Rule	IF	THEN	CF	Pri.	Reason / evidence
R4	Skin rash AND Pain behind the eyes	Dengue signs	0.85	30	Most specific dengue pair
R5	Body / muscle / joint pain AND Nausea or vomiting	Dengue signs	0.60	30	Dengue: aches + nausea/vomiting
R6	Bleeding gums or nose	Dengue signs	0.70	30	Dengue warning sign
R7	Fever AND Dengue signs	**Dengue**	0.90	30	High fever + ≥2 dengue symptoms
R8	Chills	Malaria signs	0.60	30	Chills is one of the three most common early malaria symptoms
R9	Headache AND Nausea or vomiting	Malaria signs	0.60	30	Early malaria symptoms
R10	Fever AND Malaria signs	**Malaria**	0.90	30	Fever + malaria signs
R11	Fever AND Fever lasting more than 3 days	Prolonged fever	0.90	20	Lecture heuristic: fever 3+ days → test for infection
R12	Abdominal pain AND Diarrhoea or constipation	Typhoid signs	0.75	20	Typhoid GI pattern
R13	Prolonged fever AND Typhoid signs	**Typhoid**	0.85	20	Prolonged high fever + GI symptoms
R14	Cough AND Body / muscle / joint pain	Flu signs	0.70	20	Dry cough + muscle pain
R15	Headache AND Fatigue	Flu signs	0.50	20	Headache + severe malaise
R16	Fever AND Flu signs	**Influenza**	0.85	20	Sudden fever + flu signs; cf. CDC ILI definition
R17	Runny nose AND Sneezing	Cold signs	0.80	10	Typical cold pattern
R18	Sore throat	Cold signs	0.50	10	Sore throat
R19	Cold signs	**Common Cold**	0.80	10	Most colds have no fever
R20	Sweating	Malaria signs	0.40	30	Sweating stage of the malaria paroxysm; weak evidence on its own

3.4 Red-flag rules (factual knowledge, priority 100)

Table 9. A red flag shows an emergency message as soon as its symptoms are present. It does not add a disease.

Rule	IF	Alert message	Source
RF1	Bleeding gums or nose	Bleeding gums or nose can be a warning sign of severe dengue. Seek emergency medical care now.	WHO, CDC
RF2	Chest pain AND Shortness of breath	Chest pain with shortness of breath needs urgent assessment. Seek emergency medical care now.	WHO (pneumonia)
RF3	Confusion	Confusion or reduced consciousness can signal a severe infection such as severe malaria. Seek emergency medical care now.	WHO (malaria, pneumonia)

Knowledge types: 29 factual items (6 diseases, 20 symptoms, 3 red flags) and 20 heuristic rules. Eight checks in kb_verify.py run at every start-up (known symptoms, complete diseases, valid CFs, no unreachable disease, no rule cycle, and others).

4. Explanation of Inference Reasoning

4.1 Architecture

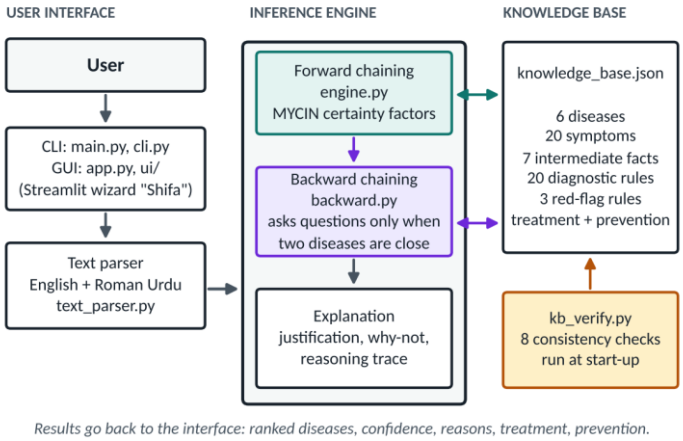


Figure 1. Architecture: knowledge base, inference engine (forward + backward), explanation, interfaces.

4.2 Forward chaining

Start from the user's facts; fire one rule per cycle until no rule can fire. Certainty factors follow MYCIN (Shortliffe and Buchanan, 1975).

Table 10. Forward-chaining method and certainty-factor formulas.

Step	How it works	Value
Working memory	Dictionary {fact: CF}; starts with the user's symptoms	—
Conflict set	Rules whose IF parts are all in memory with min CF ≥ threshold	threshold 0.2
Conflict resolution	Highest priority, then lowest rule depth, then file order	red flag 100 > severe 30 > moderate 20 > mild 10
AND of conditions	Weakest part decides	min(CFs)
Rule conclusion	Rule CF × min of its IF CFs	c = CF × min
Same fact, two rules	Evidence adds up, never above 1	new = old + c × (1 − old)
Stop	No rule left to fire (kb_verify forbids cycles, so it always stops)	fixed point
Show a disease	CF at or above the display threshold	0.30

Worked trace: Fever (moderate), Skin rash (yes), Pain behind the eyes (moderate).

Table 11. Engine trace for the dengue query (2 cycles, a 2-level chain).

Cycle	Conflict set	Fired	Calculation	New value
1	R4	**R4**	min 0.700, c = 0.595	Dengue signs: 0.000 → 0.595
2	R7	**R7**	min 0.595, c = 0.536	Dengue: 0.000 → 0.536

4.3 Severe before mild

Rules of severe diseases have priority 30 and mild ones 10, so dangerous diseases are processed first. The final list sorts by severity, then CF. Example: fever, skin rash, pain behind the eyes, runny nose, sneezing, sore throat.

Table 12. Mixed case: the cold scores higher (0.57 against 0.54), but severity decides the order.

Rank	Disease	Severity	Evidence (CF)	Confidence	Why this place
1	**Dengue**	Severe	0.54	**48.4%**	severe goes first
2	**Common Cold**	Mild	0.57	**51.6%**	mild, though CF is higher

4.4 Justification (explanation) of each diagnosis

Each diagnosis carries the rule chain down to the user's symptoms. The CLI prints it under "Why this was suggested"; the GUI shows it under "Why this?". A why-not output also explains diseases that were not shown.

```
Dengue: CF 0.536
  R7 [Fever 0.700, Dengue signs 0.595] -> min 0.595 x rule CF 0.90 = 0.536
    Dengue signs: CF 0.595
      R4 [Skin rash 1.000, Pain behind the eyes 0.700] -> min 0.700 x rule CF 0.85 = 0.595
```

4.5 Backward chaining on close calls

If the top two diseases are almost tied (gap = (A − B) / (A + B) × 100 ≤ 15 points), each becomes a goal. The engine finds the rules that conclude it, checks their IF parts and turns unknown symptoms into questions (max 3). Questions are generated from the rules, not typed by hand.

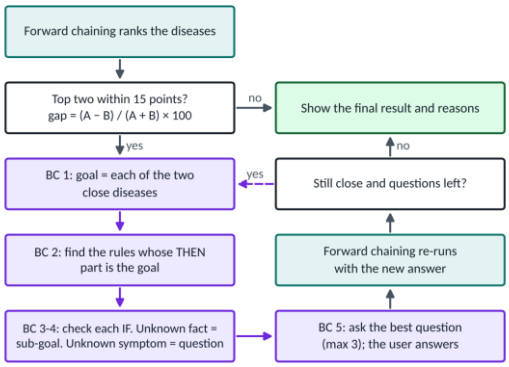


Figure 2. Hybrid reasoning: forward chaining first, backward chaining on close calls.

Table 13. Example: fever, headache, nausea, body aches (all moderate) gives Dengue 0.378 and Malaria 0.378, a 50/50 tie. What each answer does:

Answers	Questions	Dengue	Malaria	Outcome
None (tie)	0	0.378 (50.0%)	0.378 (50.0%)	tie
Bleeding gums: Yes	1	**0.630 (62.5%)**	0.378 (37.5%)	separated; RF1 alert
Bleeding gums: No, Chills: moderate	2	0.378 (38.8%)	**0.597 (61.2%)**	separated
No, No, No	3	0.378 (50.0%)	0.378 (50.0%)	still tied; limit reached

4.6 Confidence levels

Table 14. Two numbers are shown per disease.

Number	Meaning	Formula
Confidence %	Share among the diseases shown (adds up to 100%), like Disease A 62.5%, Disease B 37.5%	CF ÷ sum of shown CFs × 100
Evidence strength	How strong the evidence is on its own (raw CF, 0 to 1)	from the rules

Example: Dengue 0.630 and Malaria 0.378 sum to 1.008, so Dengue = 62.5% and Malaria = 37.5%. A single weak match would show 100%, so the evidence strength is shown next to it.

5. Example Queries and Screenshots

The system has a CLI (python main.py) and a GUI (python -m streamlit run app.py). All numbers below come from the running engine. The table below lists the five queries; the screenshots follow.

Table 15. Five example queries and their results.

#	User input	Result (ranked)	CF	Confidence	Alert / note
Q1	fever, skin rash, pain behind the eyes	Dengue (severe)	0.54	100%	no alert; forward chaining
Q2	fever, headache, nausea, body aches	Dengue vs Malaria tie; after Bleeding gums = Yes: Dengue, Malaria	0.63 / 0.38	62.5% / 37.5%	backward chaining; RF1
Q3	fever, cough, shortness of breath, chest pain	Pneumonia (severe)	0.60	100%	RF2 fires in cycle 1
Q4	fever, skin rash, pain behind eyes, runny nose, sneezing, sore throat	Dengue, then Common cold	0.54 / 0.57	48.4% / 51.6%	close call (1 question); severe listed first
Q5	fever (alone)	Not enough evidence	—	—	needs 2+ symptoms

5.1 Q1: forward chaining (dengue)

Input: fever, skin rash, pain behind the eyes. R4 proves Dengue signs (0.595), then R7 proves Dengue (0.536). Treatment shown: rest, fluids, paracetamol, no aspirin or ibuprofen, see a doctor. Prevention: repellent, screens, remove standing water.

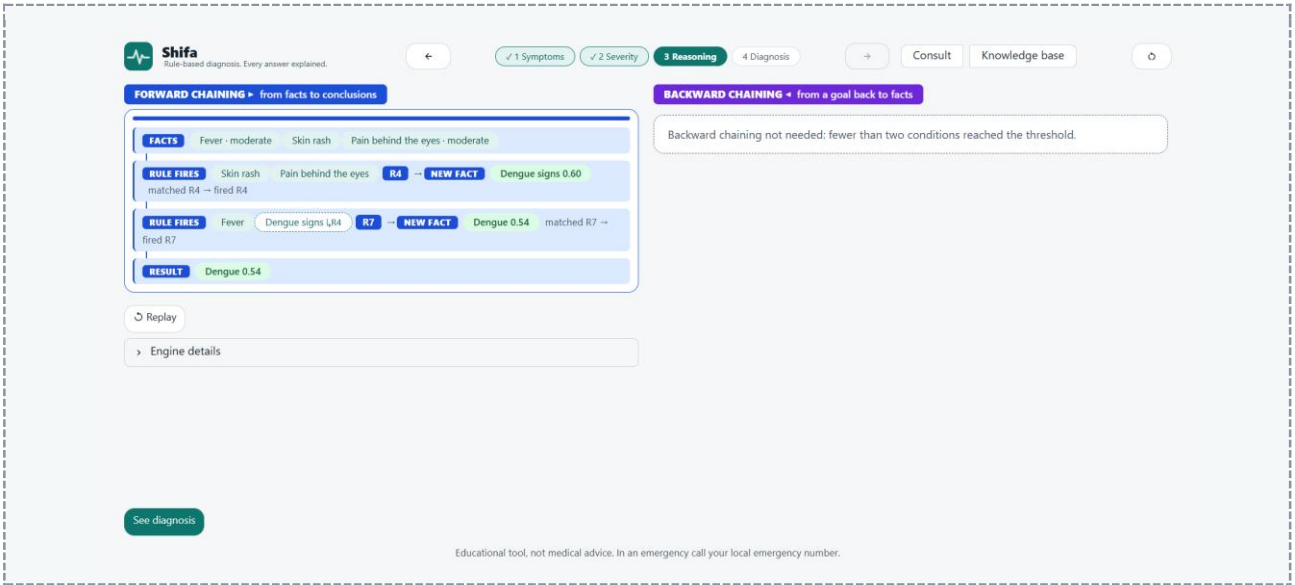


Figure 3. GUI Step 3: forward-chaining panel (FACTS, RULE FIRES, NEW FACT, RESULT).

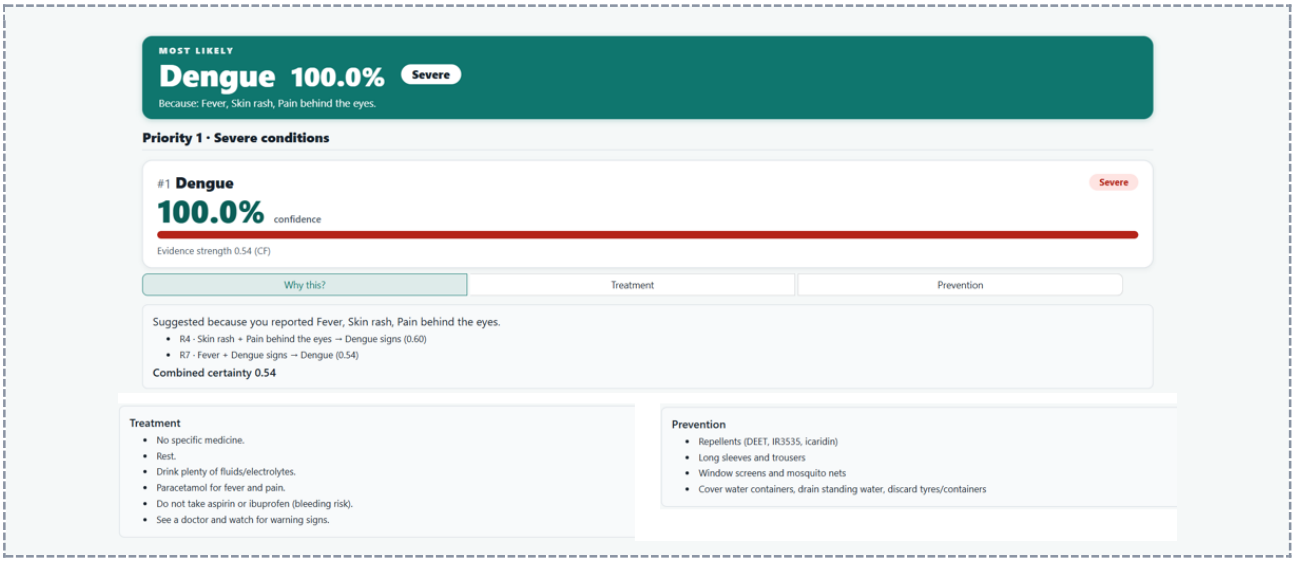


Figure 4. GUI Step 4: diagnosis with Why this?, Treatment and Prevention open.

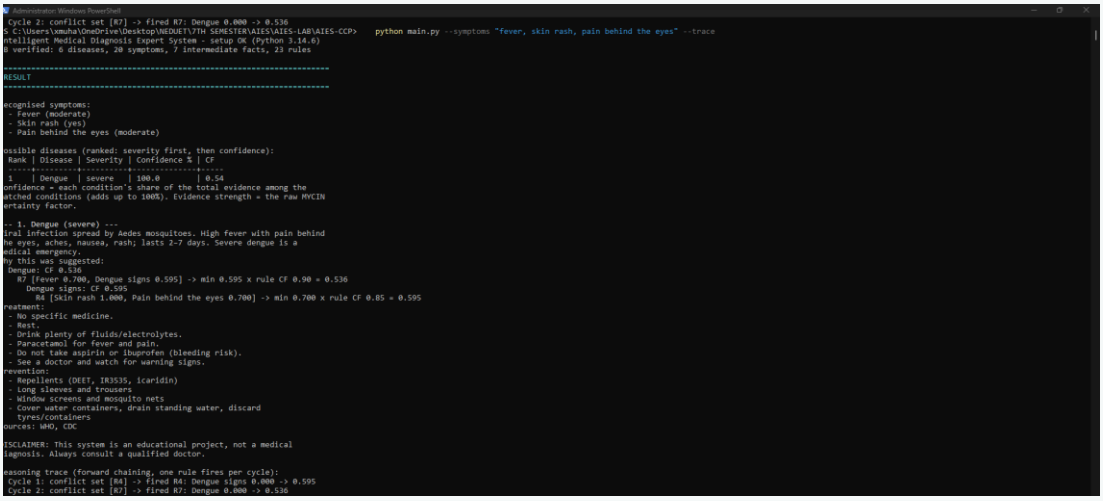


Figure 5. CLI output with the reasoning trace (Q1).

5.2 Q2: backward chaining on a close call

Input: first example button (fever, headache, nausea, body aches). Forward result is a 50/50 tie, so backward chaining asks: "Bleeding gums or nose?" (Dengue ← R7 ← R6). Answer Yes: Dengue 62.5%, Malaria 37.5%, red flag RF1. Answer No then Chills: Malaria 61.2%, Dengue 38.8%.

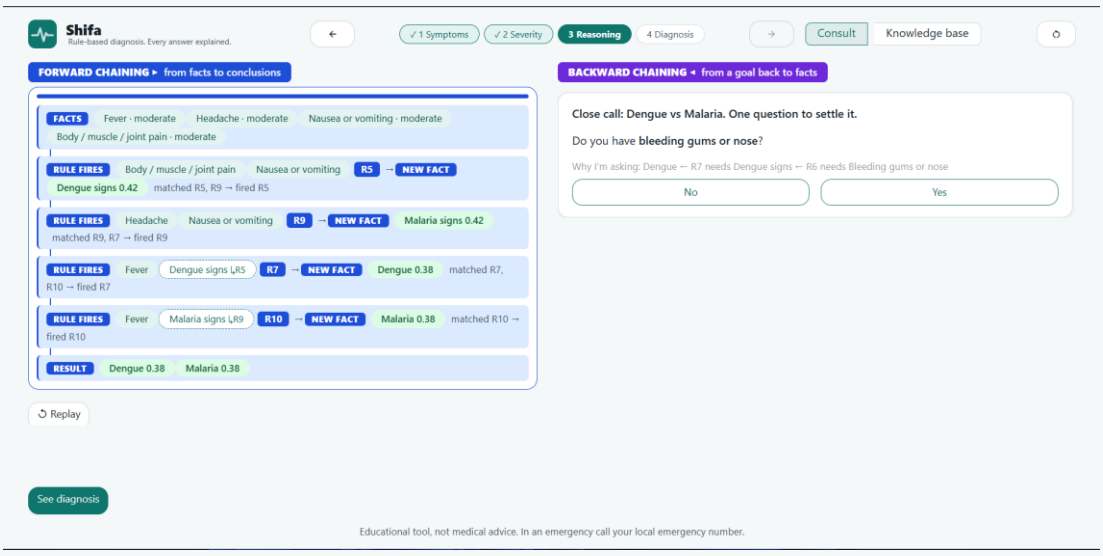


Figure 6. GUI Step 3: backward-chaining panel (violet) with the question card.



Figure 7. GUI Step 4: after answering Yes, red alert, Dengue 62.5%, Malaria 37.5%.

5.3 Q3: red-flag (emergency) warning

Input: fever, cough, shortness of breath and chest pain. RF2 has priority 100 and fires in cycle 1, before any diagnosis; then R1, R2 and R3 give Pneumonia 0.595. Without fever, the alert still appears but no disease is shown.

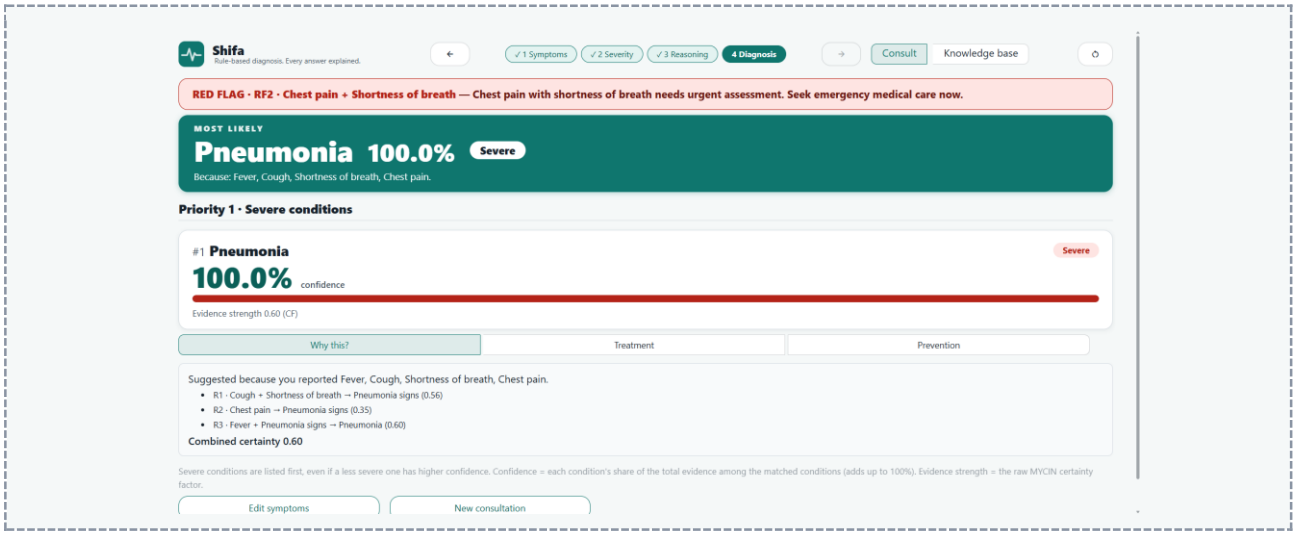


Figure 8. GUI Step 4: red-flag banner (RF2) above the Pneumonia verdict.

5.4 Q4: severe disease listed before a mild one

Input: fever, skin rash, pain behind the eyes, runny nose, sneezing, sore throat. The cold has the higher CF (0.57 against 0.54), so this is also a close call: backward chaining asks "Bleeding gums or nose?" and we answer No. Dengue is still listed first because it is severe (Section 4.3).

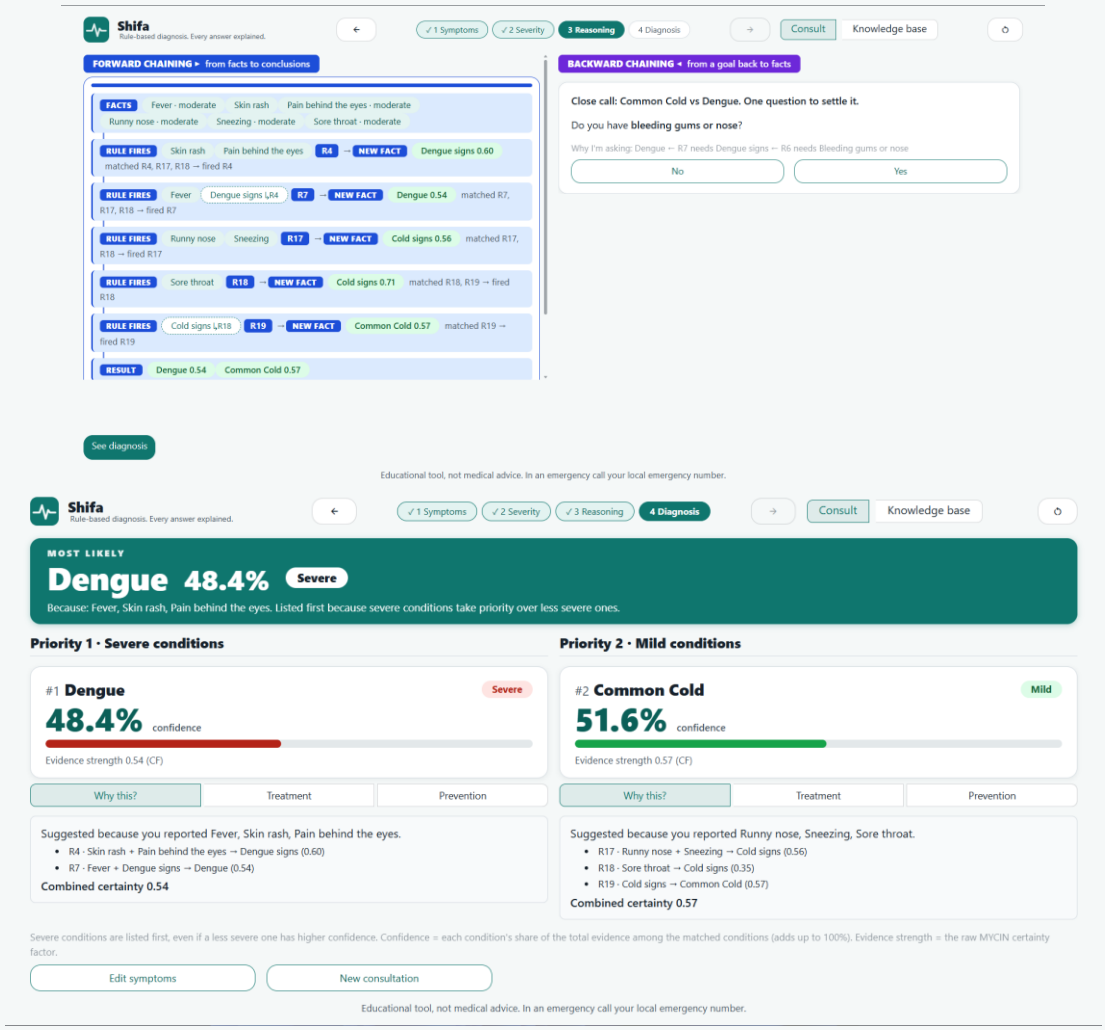


Figure 9. GUI Step 4: priority tiers, Dengue (severe) above Common cold (mild).

5.5 Q5: not enough evidence

Input: fever alone. Fever is required for five of the six diseases, but one symptom does not reach the 0.30 threshold. The system says "not enough evidence" instead of guessing. The same happens in the GUI.

6. Testing and Validation

python run_tests.py runs nine groups and prints a summary. All 132 checks pass.

Table 16. Test groups.

#	Group	Checks	What it proves
1	Knowledge-base verification	1	8 consistency checks pass (6 diseases, 20 symptoms, 7 facts, 23 rules)
2	Clinical cases T01–T14	14	WHO/CDC symptom lists give the expected disease, ranking and alerts
3	Regression A–G	7	Exact CF values stay the same (tolerance 0.001)
4	CLI input parsing	4	Separators, synonyms, "did you mean" hints
5	Verifier self-tests	3	Deliberately broken knowledge bases are caught
6	Order independence	50	50 random rule orders give identical results
7	Hybrid backward chaining	8	Close calls, generated questions, scripted answers, no repeated question
8	GUI smoke tests	31	Wizard order, replay, red-flag banner, priority tiers, knowledge-base view
9	Text parser	14	English and Roman Urdu, negation, intensity, duration, typos
	Total	**132**	**All passed**

Table 17. Clinical cases (expected disease from WHO/CDC text, and the result).

ID	Case	Result	Source	Status
T01	Dengue	Dengue 0.54	WHO WPRO dengue	**PASS**
T02	Malaria, early symptoms	Malaria 0.38	WHO malaria	**PASS**
T03	Typhoid	Malaria 0.38, Typhoid 0.45	WHO typhoid	**PASS**
T04	Influenza	Influenza 0.57	WHO influenza	**PASS**
T05	Pneumonia	Pneumonia 0.60, Malaria 0.38; alert RF2	WHO pneumonia	**PASS**
T06	Common cold	Common Cold 0.52	CDC common cold	**PASS**
T07	Mixed severe + mild (severe ranked first)	Dengue 0.54, Common Cold 0.57	CCP requirement	**PASS**
T08	Edge: fever only	no disease shown	Edge case	**PASS**
T09	Edge: no symptoms	no disease shown	Edge case	**PASS**
T10	Edge: unknown symptom	no disease shown	Edge case	**PASS**
T11	Dengue warning sign	Dengue 0.74; alert RF1	CDC dengue	**PASS**
T12	Severe malaria	Malaria 0.52; alert RF3	WHO malaria	**PASS**
T13	Edge: weak evidence near the threshold	no disease shown	Edge case	**PASS**
T14	Edge: many severe symptoms but no fever	no disease shown; alert RF1+RF2+RF3	Edge case	**PASS**



Figure 10. Terminal: kb_verify and run_tests.py, TOTAL 132/132 passed.

7. Assumptions and Limitations

Medical disclaimer: educational project, not a medical diagnosis. Consult a qualified doctor.

Table 18. Assumptions.

#	Assumption
1	Certainty factors are values chosen by the group from WHO/CDC descriptions, not clinically validated probabilities.
2	The patient has one of the six diseases in the knowledge base (confidence % is a share among them).
3	Severity is fixed per disease; intensity (mild / moderate / severe) is chosen by the user, default moderate.
4	Only confirming evidence is modelled; an absent symptom never lowers a CF.
5	"Fever for more than 3 days" is asked only after fever is reported.
6	Symptoms not in the knowledge base are reported to the user and ignored.

Table 19. Main limitations (full list of 32 in docs/LIMITATIONS.md).

ID	Limitation	How it is handled	Possible fix
L01	Fever is required for 5 of 6 diseases; cases without fever (e.g. elderly pneumonia) are missed.	Follows WHO/CDC text; the system reports that fever is missing.	Add rules without fever after doctor review.
L02	Dengue with moderate fever cannot exceed CF 0.63.	MYCIN: a conclusion is never stronger than its weakest part.	Review intensity values.
L03	CFs are expert-assigned, not learned.	They rank evidence; they are not probabilities.	Calibrate with clinician or data.
L04	No interview with a doctor.	WHO/CDC sources shown on every rule.	Medical review.
L05	Confidence % is a share; one weak match shows 100%.	Evidence strength shown next to it.	Warn when best CF is low.
L11	Free text is keyword matching; English and Roman Urdu only.	Standard library; every reading is traceable.	More synonyms; Urdu script.
L17	Very little evidence gives "not enough evidence".	Refusal is safer than a guess; hint names what is missing.	Targeted questions.
L29	Layout tuned for 1366×768 and larger, not phones.	Demo on laptop or projector.	Single-column layout.

8. Conclusion and Future Work

Shifa meets every requirement of the CCP: a knowledge base of 20 symptoms, 6 diseases and 23 rules with severity, treatments and prevention; a forward-chaining engine with certainty factors, severe-first ranking, per-diagnosis justification and confidence levels; backward chaining for close calls; a CLI and a GUI; and 132 passing tests. Keeping knowledge in data and reasoning in code made the system easy to test and explain. Future work: doctor review and CF calibration, rules for cases without fever, negative evidence, more diseases (JSON only), Urdu script input.

References and Appendix: Python Source Code

References

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[4] WHO. Pneumonia health topic. who.int/health-topics/pneumonia

[5] WHO Western Pacific. Dengue and severe dengue. who.int/westernpacific/health-topics/dengue-and-severe-dengue

[6] CDC. Dengue symptoms and treatment. cdc.gov/dengue

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[9] Russell, S. and Norvig, P. Artificial Intelligence: A Modern Approach, 2nd ed., Sec. 7.5 and 14.7.

[10] Course lecture notes, CT-361 AIES: Expert Systems, Fundamentals (NED University, Fall 2026).

Appendix: Python source code (in the submission zip)

Table 20. Project files.

File	Purpose
`main.py`	Entry point: verifies the knowledge base, starts the CLI
`app.py, ui/theme.py, ui/components.py`	Streamlit GUI "Shifa" (entry point, theme, render functions)
`expert_system/engine.py`	Forward chaining with certainty factors (386 lines, no medical names)
`expert_system/backward.py`	Backward chaining: question generation (242 lines)
`expert_system/cli.py`	Command-line interface (574 lines)
`expert_system/text_parser.py`	English / Roman Urdu text to symptoms (180 lines)
`expert_system/kb_verify.py`	The 8 knowledge-base checks (240 lines)
`data/knowledge_base.json`	All medical knowledge
`run_tests.py, tests/*.json`	Test runner and test data (132 checks)
`docs/kb_design.md, docs/LIMITATIONS.md, README.md`	Design, limitations, instructions

Table 21. How to run (Python 3.11+ for the GUI).

Command	Result
`python main.py`	Interactive CLI
`python main.py --symptoms "fever, skin rash" --trace`	One query with trace
`pip install -r requirements.txt`	Installs Streamlit (GUI only)
`python -m streamlit run app.py`	GUI in the browser (offline)
`python run_tests.py`	All 132 checks